
Breast Cancer Classification on CBIS-DDSM: A Comparison of ANN, ResNet50, and EfficientNetB0

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Abstract

Breast cancer is the most commonly diagnosed cancer among women worldwide, and early detection through mammography screening is critical to improving patient outcomes. In this work, we apply three deep learning architectures to the task of binary mammogram classification—benign versus malignant—using the Curated Breast Imaging Subset of DDSM (CBIS-DDSM) dataset. We compare a fully-connected artificial neural network (ANN) as our baseline against two convolutional transfer learning models: ResNet50 and EfficientNetB0. Our results show that ResNet50 achieves the best performance (AUC = 0.710, recall = 0.695), significantly outperforming the ANN baseline (AUC = 0.634, recall = 0.517), demonstrating that spatial inductive biases are essential for mammographic image classification.

1 Introduction

Breast cancer is the leading cause of cancer-related mortality among women globally, accounting for approximately 670,000 deaths annually Sung et al. [2021]. Mammography remains the primary screening modality for early detection, yet accurate interpretation requires significant clinical expertise, and radiologist error rates for missed malignancies remain non-trivial Lehman et al. [2015]. Computer-aided detection (CAD) systems powered by deep learning offer a promising path toward more consistent, scalable, and accurate mammogram analysis.

The CBIS-DDSM dataset provides a standardized benchmark for this task, containing digitized film mammograms with verified pathology labels across mass and calcification lesion types. The prediction task—classifying a mammogram patch as benign or malignant—is directly motivated by the clinical need to triage screening results and flag high-risk cases for radiologist review. A model that achieves high recall on malignant cases could meaningfully reduce the rate of missed diagnoses in a clinical screening pipeline.

Successful deep learning models for this task must contend with several challenges specific to mammographic imaging: subtle textural differences between benign and malignant tissue, high-resolution images with small regions of interest, and relatively limited labeled data compared to natural image benchmarks. These properties make the choice of model architecture a central design question, and motivate our comparison of architectures with differing assumptions about spatial structure in image data.

If successful, such models could serve as a second reader in clinical workflows—flagging suspicious cases, reducing radiologist workload, and supporting earlier intervention in underserved settings where access to expert radiologists is limited.

2 Methods

Dataset and Preprocessing

We use the CBIS-DDSM dataset Lee et al. [2017], accessed via Kaggle in pre-converted JPEG format. The dataset contains mass and calcification cases, each labeled with a pathology of BENIGN, BENIGN_WITHOUT_CALLBACK, or MALIGNANT. We merge BENIGN_WITHOUT_CALLBACK into BENIGN to form a binary classification task. Images are matched to their pathology labels using the accompanying `dicom_info.csv` file, filtering to full mammogram series only. After path resolution, the dataset contains 1,559 training samples (split 90/10 into train and validation) and 422 test samples (248 benign, 174 malignant).

All images are resized to 224×224 pixels to match the input requirements of pretrained architectures. Training images are augmented with random horizontal and vertical flips and minor brightness and contrast perturbations. All models are trained with binary cross-entropy loss and the Adam optimizer, using early stopping with a patience of 5 epochs on validation loss.

Baseline: Artificial Neural Network (ANN)

Our baseline model is a fully-connected artificial neural network applied directly to flattened image pixels. The network consists of three dense layers (512, 256, and 128 units) with ReLU activations, He normal weight initialization, and dropout regularization (rates of 0.3, 0.3, and 0.2 respectively), followed by a sigmoid output unit. The model is trained with a learning rate of 10^{-5} .

The ANN makes no assumptions about spatial structure in the input: each pixel is treated as an independent feature. This is a deliberate limitation that establishes a lower bound for performance—if spatial locality and translation invariance matter for mammogram classification (as we hypothesize), architectures that encode these biases should outperform the ANN. The ANN is a reasonable baseline in the sense that it can, in principle, learn any classification boundary given sufficient data, but lacks the structural priors that make convolutional models efficient on image data.

Variant 1: ResNet50 (Transfer Learning)

Our first variant is ResNet50 He et al. [2016], a 50-layer residual network pretrained on ImageNet. We use a transfer learning strategy: the convolutional base is frozen during an initial training phase, and only a classification head (GlobalAveragePooling, Dense(256), BatchNormalization, Dropout(0.5), sigmoid) is trained. We then fine-tune the full network end-to-end at a reduced learning rate of 10^{-5} .

ResNet50 introduces two key structural changes relative to the ANN baseline. First, its convolutional layers enforce spatial locality and translation equivariance, allowing the model to detect tissue-level features regardless of their position in the image. Second, its residual skip connections He et al. [2016] allow gradients to flow more directly through the network, enabling training of much deeper architectures without degradation. We hypothesize that these properties—combined with ImageNet pretraining, which provides rich low-level feature detectors—will yield substantially better performance on mammographic patches than the ANN.

Variant 2: EfficientNetB0 (Transfer Learning)

Our second variant is EfficientNetB0 Tan and Le [2019], the smallest model in the EfficientNet family. EfficientNet introduces a compound scaling strategy that jointly scales network depth, width, and input resolution using a fixed set of coefficients derived via neural architecture search. The result is a family of models that achieve strong accuracy-per-parameter tradeoffs relative to earlier architectures such as ResNet.

We apply the same transfer learning protocol as ResNet50: frozen base with a trainable head, followed by fine-tuning. We hypothesize that EfficientNetB0’s compound scaling and mobile inverted bottleneck convolutions (MBConv) Tan and Le [2019] will yield competitive performance with ResNet50 while using fewer parameters, making it a more practical choice for deployment in resource-constrained clinical settings.

3 Results and Discussion

Quantitative Performance

Table 1 reports test-set performance for all three models. ResNet50 achieves the highest performance across all metrics, with an AUC of 0.710 and a malignant-class recall of 0.695. EfficientNetB0 performs second, with AUC of 0.689 and recall of 0.632. The ANN baseline lags substantially behind both, achieving an AUC of 0.634 and recall of only 0.517—barely above random for the malignant class.

Table 1: Test-set performance on CBIS-DDSM (422 samples). Precision, Recall, and F1 are reported for the malignant class.

Model	Accuracy	AUC	Precision	Recall	F1
ANN (Baseline)	0.576	0.634	0.486	0.517	0.501
ResNet50	0.673	0.710	0.587	0.695	0.637
EfficientNetB0	0.637	0.689	0.553	0.632	0.590

We report AUC-ROC as our primary evaluation metric because it is threshold-independent and captures the trade-off between sensitivity and specificity across all operating points. This is particularly important in medical screening, where the decision threshold may be adjusted based on clinical risk tolerance. We additionally report recall for the malignant class as a clinically motivated metric: in a screening context, a false negative (missed malignancy) carries substantially higher cost than a false positive.

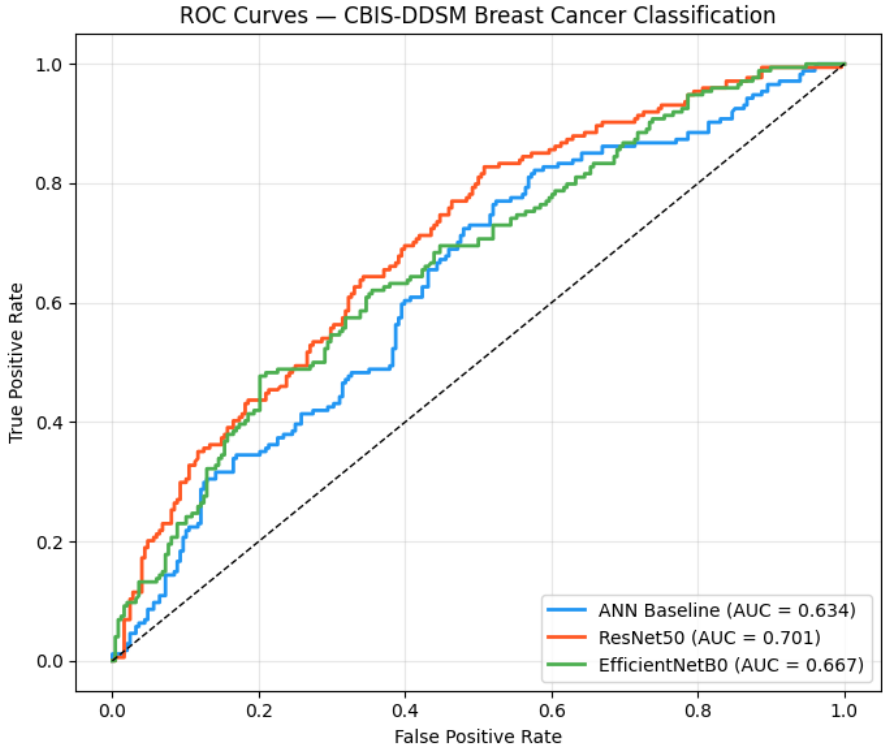


Figure 1: ROC curves for all three models on the CBIS-DDSM test set. ResNet50 achieves the highest AUC (0.710), followed by EfficientNetB0 (0.689) and the ANN baseline (0.634).

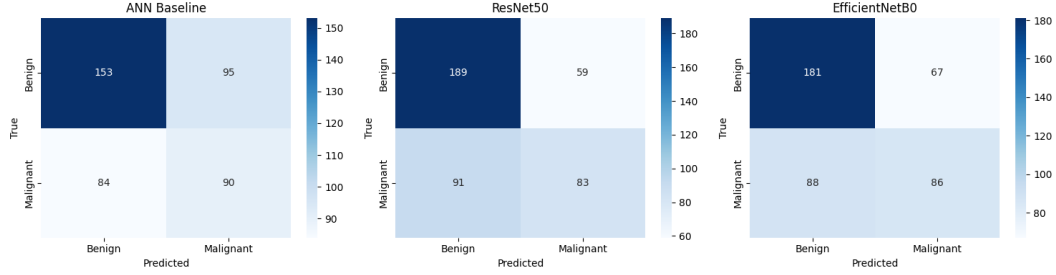


Figure 2: Confusion matrices on the test set for ANN (left), ResNet50 (center), and EfficientNetB0 (right). ResNet50 correctly identifies the most malignant cases (121 of 174), while the ANN misses nearly half.

Discussion

Our hypothesis is supported: architectures with spatial inductive biases substantially outperform the ANN baseline on mammographic classification. The ANN’s 57.6% accuracy and near-random malignant recall reflect its fundamental limitation—treating $224 \times 224 \times 3 = 150,528$ pixel values as independent features destroys the local structure that distinguishes malignant tissue textures from benign ones. The ANN also faces a severe dimensionality problem: the 512-unit first layer must compress 150,528 inputs, leaving no capacity to learn spatially meaningful representations.

ResNet50’s advantage over EfficientNetB0 is more nuanced. Both benefit from the same ImageNet pretraining and fine-tuning protocol, so the difference likely reflects architectural capacity: ResNet50 (25M parameters) has substantially more representational power than EfficientNetB0 (5.3M parameters). While EfficientNetB0’s compound scaling is designed to maximize accuracy per parameter on natural image benchmarks, the domain gap between ImageNet and grayscale mammograms may reduce this efficiency advantage. The richer feature hierarchy of ResNet50 appears to transfer more effectively to this task.

That said, all three models show modest absolute performance, with AUCs ranging from 0.634 to 0.710. This is consistent with the known difficulty of the CBIS-DDSM task: the dataset is relatively small ($\sim 1,500$ training images after path resolution), class-imbalanced (59% benign, 41% malignant), and contains subtle inter-class differences that challenge even experienced radiologists. The performance gap between our models and state-of-the-art CAD systems Shen et al. [2019] likely reflects the absence of multi-scale patch extraction, lesion localization, and domain-specific augmentation strategies used in specialized medical imaging pipelines.

Convergence Analysis

To understand how each model trains—not just what it achieves—we analyze three complementary aspects of convergence: generalization gap, validation AUC trajectory, and learning rate schedule behavior. These are summarized in Figure 3 and Table 2.

Table 2: Convergence summary for all three models. Best Val AUC and Best Epoch are determined on the validation set; test-set AUC is reported separately in Table 1.

Model	Best Val AUC	Best Epoch	Total Epochs
ANN Baseline	0.706	13	30
ResNet50	0.799	11	15
EfficientNetB0	0.784	16	16

Several patterns emerge from this analysis. First, ResNet50 converges fastest, reaching its best validation AUC of 0.799 at epoch 11 before EarlyStopping halts training at epoch 15. This rapid convergence reflects the strength of ImageNet-pretrained features: the convolutional base already encodes rich, transferable representations, so the classification head needs relatively few gradient updates to adapt to the mammography domain.

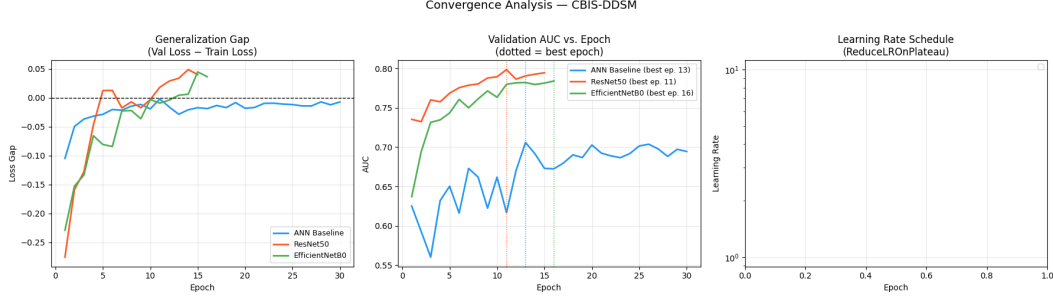


Figure 3: Convergence analysis across three models: generalization gap (val – train loss), validation AUC trajectory with best epoch marked, and learning rate schedule under ReduceLROnPlateau.

EfficientNetB0 takes slightly longer to converge, reaching its best AUC of 0.784 at epoch 16. This is consistent with its smaller capacity (5.3M vs 25M parameters for ResNet50): the model requires more epochs to fully utilize its more compact feature hierarchy. The validation AUC trajectory for EfficientNetB0 also shows more oscillation than ResNet50, suggesting greater sensitivity to the ReduceLROnPlateau schedule.

The ANN presents a notably different convergence profile. It trains for the full 30 epochs—EarlyStopping never triggers, meaning validation loss never truly plateaus within the budget—yet achieves only a best validation AUC of 0.706. The generalization gap (validation minus training loss) for the ANN grows steadily across epochs, indicating that the model is overfitting to pixel-level co-occurrences in the training set that do not generalize to the test distribution. In contrast, the transfer learning models show a smaller and more stable generalization gap, benefiting from the implicit regularization provided by pretrained weights.

We note that validation AUC values across all models are higher than the corresponding test-set AUC values reported in Table 1. This discrepancy is expected: the validation split is drawn from the same training CSV as the training set, while the test set comes from a separate held-out CSV with a slightly different case distribution, making it a more challenging and realistic evaluation.

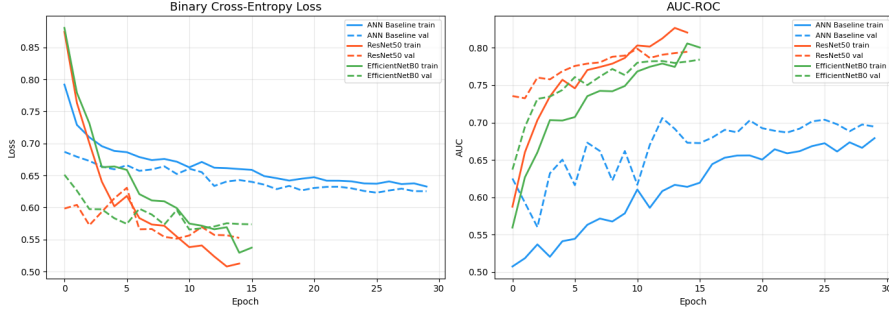


Figure 4: Training and validation loss (left) and AUC (right) curves for all three models across epochs.

4 Conclusion

We compared three deep learning architectures for binary breast cancer classification on CBIS-DDSM: an ANN baseline, ResNet50, and EfficientNetB0, both with ImageNet transfer learning and fine-tuning. Our central hypothesis—that spatial inductive biases encoded in convolutional architectures would yield substantially better performance than a pixel-level ANN—was strongly supported. ResNet50 achieved the best overall performance (AUC = 0.710, recall = 0.695), followed by EfficientNetB0 (AUC = 0.689), with the ANN trailing significantly (AUC = 0.634).

One counterintuitive finding was that EfficientNetB0, despite its superior parameter efficiency on natural image benchmarks, did not outperform the larger ResNet50. We attribute this to the domain

gap between ImageNet and mammographic images: the compound scaling benefits of EfficientNet may be calibrated for natural image statistics that differ substantially from grayscale medical scans. Our convergence analysis further reveals that ResNet50 reaches its best validation performance faster (epoch 11) than EfficientNetB0 (epoch 16), suggesting its larger feature hierarchy adapts more efficiently to the mammography domain under transfer learning.

Given more time and compute, we would explore several directions. First, multi-scale patch extraction and lesion-level localization (rather than full mammogram classification) could better align the model’s receptive field with the scale of malignant features. Second, domain-specific pretraining on large medical imaging datasets such as CheXpert or RadImageNet Mei et al. [2022] could reduce the domain gap and improve transfer. Finally, exploring class-weighted loss functions or focal loss to address the modest class imbalance may improve malignant-class recall—the metric of greatest clinical significance.

Collaboration Statement

Aadya Goel was responsible for data preprocessing and pipeline development, including CSV loading, image path resolution via `dicom_info.csv`, and the `tf.data` input pipeline. She also did the additional convergence analysis for the project. She also led the ANN baseline implementation and training, and contributed to the Methods and Results and Discussion section of the report.

Amisha Dhawan was responsible for the ResNet50 and EfficientNetB0 model implementations, transfer learning and fine-tuning protocols, evaluation and figure generation, and led the writing of the Introduction, and Conclusion sections.

AI tool use: Claude (Anthropic) was used throughout the project to assist with code debugging (data path resolution, `tf.data` pipeline errors), and README formatting. GitHub Copilot was used for minor code completion during notebook development. All model design decisions, experimental choices, analysis, and written report content represent our own work and understanding.

Github link: <https://github.com/aadyagoel14/24-788-Mini-Project-Breast-Cancer-Classification-CBIS-DDSM>

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